



**Maryland Comprehensive
Assessment Program**

**Grade 8
MISA
Practice Test**



Maryland
STATE DEPARTMENT OF EDUCATION

Section 1

Directions:

Today, you will take Section 1 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

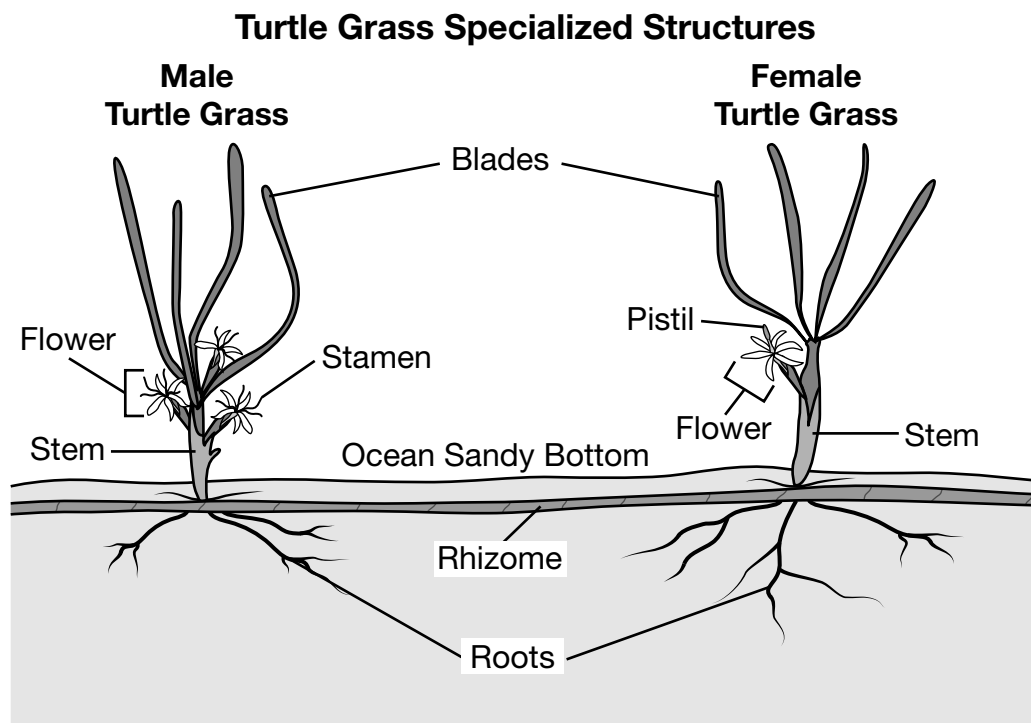
Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

Turtle grass, a type of seagrass, is a flowering plant that lives in shallow coastal areas. Turtle grass is an important component in ocean ecosystems because it filters out pollutants, cycles nutrients, and provides shelter to small marine organisms and their offspring. An area measuring 0.0040 square kilometer of turtle grass can support more than forty thousand fish and fifty million small marine organisms.

- **asexually, by using a horizontal stem called a rhizome and a root system under the ocean sandy bottom to quickly establish a community**
- **sexually, by using ocean currents that carry pollen, a method that can take longer to establish a community than asexual reproduction**
- **sexually, by relying on other marine organisms serving as pollinators, which has been shown to speed up reproduction compared to the ocean current method of sexual reproduction**

Scientists determined that there are male and female turtle grass plants with specialized structures.



To better understand the three reproductive methods of turtle grass, scientists recorded underwater videos for several days and nights.

The videos showed the flowers of male plants opening at night. Small marine organisms and fish were selectively feeding on the pollen from the male plants. The videos also showed pollen sticking to the bodies of small marine organisms and fish after they interacted with the male plants. Small marine organisms would later get briefly stuck to the female's pistil. This allowed the pollen to be deposited, which could lead to a seed forming.

In addition, the pollen that was eaten by small fish would be carried large distances before being deposited in their waste. In some instances, this pollen would come into contact with a female plant, leading to fertilization and the growth of a new plant.

The scientists observed that this method of sexual reproduction led to seagrass populations that covered large amounts of space. The scientists determined that the small marine organisms and fish were turtle grass pollinators. The scientists called them the "Bees of the Seas."

Turtle Grass Traits

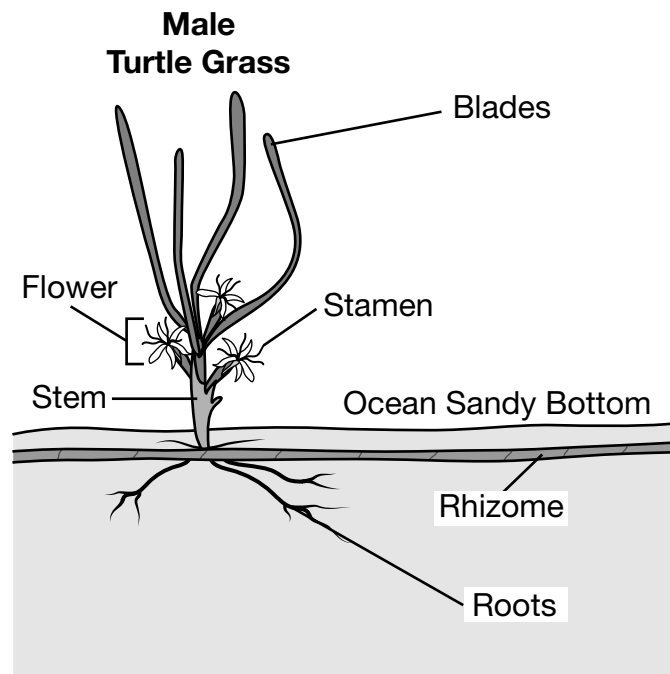
Traits	Average Range	Maximum Measurement	Minimum Measurement
Dry Mass (g/m ²)	300-600	1500	100
Blade Length (cm)	10-20	35	4
Blade Width (mm)	10	38	6

The scientists discovered that some environmental factors can affect these traits, as shown in the table.

Turtle Grass Information

Factors	Effect on Population Size	Effects on Traits
Decrease in water clarity	Large decrease in population numbers	• Blade length: large number of blade lengths measuring 4 cm • Blade width: large number of blade widths measuring 6 mm
Increase in rainfall	Small decrease in population numbers	• Average dry mass: large number of plant parts measuring 100 g/m^2 • Blade length: large number of blade lengths measuring 4 cm • Blade width: large number of blade widths measuring 6 mm
Increase in fast-moving underwater currents	Small decrease in population numbers	• Blades were broken off, causing large number of blade fragments shorter than 4 cm

- 2 Scientists claim that turtle grass plants are successful at reproducing sexually.



Which two specialized structures support this claim?

- Ⓐ blades
- Ⓑ flower
- Ⓒ stamen
- Ⓓ stem
- Ⓔ rhizome

4 Based on the evidence, an increase in the erosion of the ocean's sandy bottom would most directly affect the

-

5 Based on the evidence, which pair includes an environmental factor that will have a positive effect on turtle grass growth and an environmental factor that will have a negative effect on turtle grass growth?

- Ⓐ a storm that produces large waves and an increase in water depth
- Ⓑ a large number of sunny days and an increase in water depth
- Ⓒ a storm that produces large waves and more runoff
- Ⓓ a large number of sunny days and less runoff

Write your response on the lines provided.

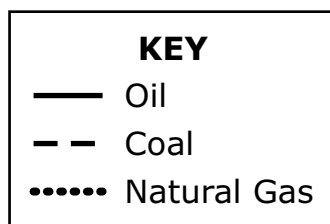
This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

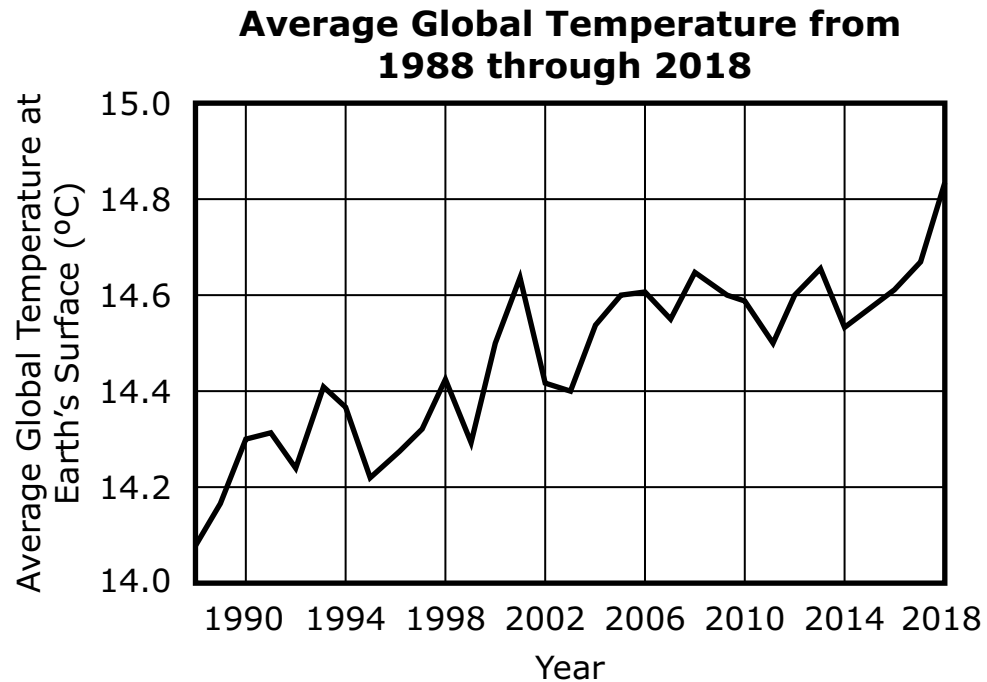
A scientist researched an island in the Chesapeake Bay. The land area of the island had decreased over time. In 2018, the island was completely under seawater. Rising seawater levels and erosion from storms partially caused the changes to the island.

The scientist studied the relationship between the global energy use from fossil fuels, the average global temperature at Earth's surface, and the rise in seawater levels around the island. Global energy is measured in zettajoules (ZJ). Average global temperature is measured in degrees Celsius (°C). The graphs show the data the scientist collected.

The graph displays three data series: Total (solid line), Industrial (dashed line), and Residential (dotted line). The Y-axis represents Global Energy Used in ZJ, ranging from 50,000 to 200,000. The X-axis represents the Year, from 1990 to 2018. All three series show a general upward trend over the period.

Year	Total (ZJ)	Industrial (ZJ)	Residential (ZJ)
1990	130,000	90,000	65,000
1992	135,000	90,000	70,000
1994	135,000	90,000	70,000
1996	140,000	95,000	75,000
1998	150,000	95,000	80,000
2000	155,000	100,000	85,000
2002	160,000	110,000	90,000
2004	165,000	135,000	95,000
2006	170,000	145,000	100,000
2008	170,000	145,000	105,000
2010	175,000	155,000	110,000
2012	175,000	160,000	115,000
2014	180,000	160,000	120,000
2016	190,000	155,000	125,000
2018	195,000	155,000	135,000

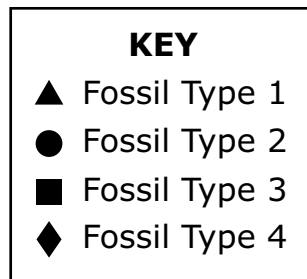




The scientist also studied data of the overall amounts of greenhouse gases in Earth's atmosphere. The data was from 1988 through 2018. The amount of carbon dioxide increased by nearly 16%. The amount of methane increased by nearly 9%.

The scientist started by recording characteristics of an outcrop found in the mountains of Maryland. The outcrop contained:

- ## Outcrop Characteristics



7 Which question can be answered based on evidence from the data?

- Ⓐ Does the global energy used by humans from fossil fuels follow the same overall trend as the average global temperature at Earth's surface from 1900 through 2000?
- Ⓑ Does the global energy used by humans from oil follow the same overall trend as the average global temperature at Earth's surface from 1988 through 2018?
- Ⓒ Does the rate of change of the global energy used by humans from coal follow the same overall trend as the rate of change in the rise in seawater levels from 2000 through 2016?
- Ⓓ Does the rate of change of the global energy used by humans from natural gas follow the same overall trend as the rate of change in the rise in seawater levels from 2010 through 2014?

8 The scientist asked this question:

Is an increase in the global energy used by humans from fossil fuels related to an increase in the global amounts of greenhouse gases in Earth's atmosphere?

Which two pieces of evidence from the data can be used to answer the scientist's question?

- Ⓐ The global energy used by humans from oil changed from 1988 through 2018.
- Ⓑ The global energy used by humans from coal changed from 2014 through 2016.
- Ⓒ The global energy used by humans from natural gas changed from 2008 through 2009.
- Ⓓ The average global temperature at Earth's surface changed from 2008 through 2009.
- Ⓔ The global amounts of carbon dioxide and methane changed from 1988 through 2018.

9 Which statement best explains how the fossil record of the outcrop can be used to organize Earth's geologic history?

- Ⓐ Fossil types 1 and 2 are the oldest fossil types because they were found in the shallowest layers of the outcrop.
- Ⓑ Fossil types 1 and 3 are the oldest fossil types because they were found in the most layers of the outcrop.
- Ⓒ Fossil types 2 and 4 are the oldest fossil types because they were found in the deepest layers of the outcrop.
- Ⓓ Fossil types 3 and 4 are the oldest fossil types because they were found in the largest layers of the outcrop.

10 Which statement provides the best evidence that a volcanic eruption occurred at the outcrop?

- Ⓐ Fossil Type 4 was found in Layer Z only.
- Ⓑ A fault line is present in layers Y and Z.
- Ⓒ Igneous rock pushed into layers Y and Z.
- Ⓓ Layers V through Z do not share a common fossil.

- 11 The scientist wanted to explain that an organism identified at the outcrop most likely became extinct before all other organisms.**

Which statement best supports the explanation?

- Ⓐ Fossil Type 3 was not found in Layer V, and the organism became extinct before Layer V formed.
- Ⓑ Fossil Type 4 was not found in layers V through Y, and the organism became extinct before layers V through Y formed.
- Ⓒ Fossil Type 2 was not found in layers V and Z, and the organism became extinct after Layer Z formed but before Layer V formed.
- Ⓓ Fossil Type 1 was not found in layers Y and Z, and the organism became extinct due to contact with liquid rock from Earth's hot interior.

Write your response on the lines provided.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Section 2

Directions:

Today, you will take Section 2 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

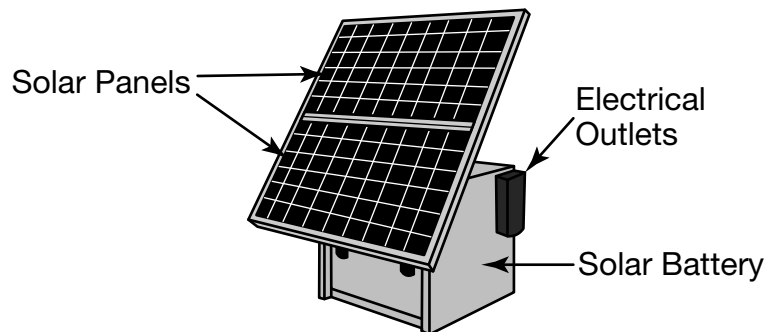
As part of a competition, a middle school engineering club was given the design challenge of providing electricity to communities where less than 10% of the population has reliable access to electricity. The research included data that compared solar- and gasoline-powered generators, which are frequently used to provide power to remote locations.

ELECTRICAL POWER SOURCES

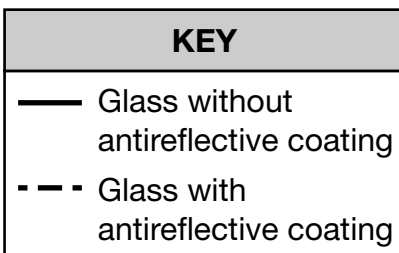
Criterion	Solar Generator	Gasoline Generator
Initial cost	\$2,000 to \$4,000	\$800 to \$1,000
Ongoing cost per year	\$30 to \$40	\$200 to \$300
25-year lifetime cost of generator	\$3,000 to \$5,000	\$6,000 to \$8,500
Operating sound level in decibels (dB) at a distance of 6 m	0	50 to 70
Environmental effect	• Uses renewable energy • No emissions	• Uses nonrenewable fuel • Emissions
Power availability	Depends on sunlight	Continuous and on demand

The members of the club decided to develop a device that used a renewable form of energy, so they completed additional research on solar panels. Solar panels are flat and covered with glass. Under the glass is a material that can convert energy from the sun into electricity.

PORTABLE SOLAR GENERATOR



ANTIREFLECTIVE COATING EFFECT



- 1 Which statement best describes the design problem that the engineering club needs to address?**
- Ⓐ New types of batteries that are less expensive and lighter need to be developed.
 - Ⓑ Local and national governments need to be encouraged to invest in solar power infrastructure.
 - Ⓒ Locations without electricity would benefit from convenient, inexpensive, and environmentally friendly power sources.
 - Ⓓ The majority of the electricity produced on Earth is from nonrenewable sources, and new technologies need to be developed to utilize renewable sources.
- 2 Which design criteria must be met to achieve the goal of the competition?**
- Ⓐ simple to operate
 - Ⓑ uses the latest technology
 - Ⓒ easy to manufacture in a local company
 - Ⓓ produces enough electrical power for a remote community
- 3 Which two criteria of the generator can be quantitatively measured?**
- Ⓐ the ease of operation
 - Ⓑ the cost of each panel
 - Ⓒ the electrical output of each panel
 - Ⓓ the material from which each panel is made
 - Ⓔ the type of antireflective coating used on each panel

- Ⓐ the initial cost of gasoline generators and the environmental effect of solar generators
- Ⓑ the lifetime cost of gasoline generators and the environmental effect of solar generators
- Ⓒ the sound level of gasoline generators and the power availability of solar generators
- Ⓓ the initial cost of gasoline generators and the power availability of solar generators

Ⓐ all of the energy because energy is always conserved

Ⓑ all of the energy because light can completely pass through glass

Ⓒ some of the energy because light is reflected off the glass

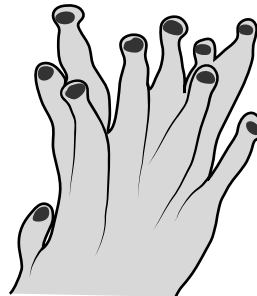
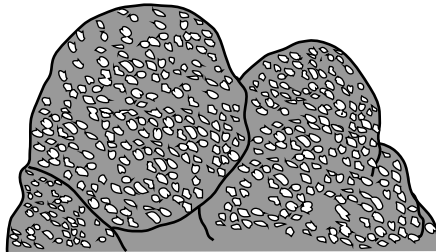
Ⓓ some of the energy because light is an electromagnetic wave

- Write your response on the lines provided.**

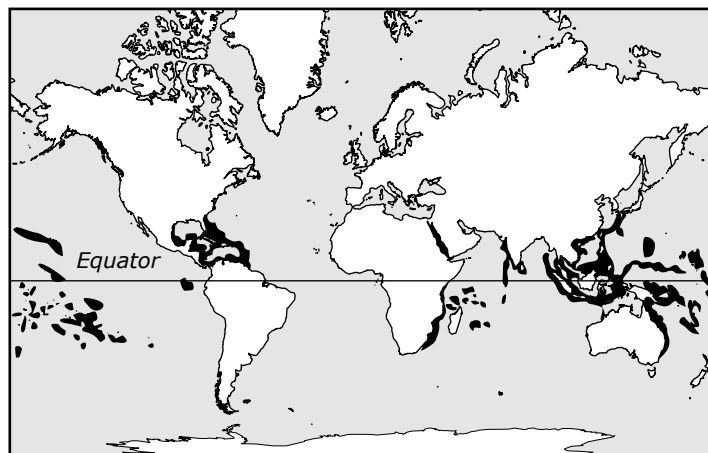
[illegible]

After a school trip to an aquarium, students from a science class researched some of the organisms that live in coral reef ecosystems. The students' research indicated that coral reefs are made up of small organisms called corals and each individual coral is called a polyp. Corals secrete calcium carbonate to form an exoskeleton, and as the corals die off, these exoskeletons build up and form a limestone foundation. New corals attach themselves to the limestone foundation and slowly form coral reefs. The following diagram shows a colony of corals and a coral polyp.

Coral Polyp



The students' research indicated that corals are animals that consume a variety of other small organisms. Corals are able to reproduce both sexually and asexually. The corals use asexual reproduction to expand colonies and use sexual reproduction to form new colonies that can be far away from the parents. Stony corals and other coral species that build reefs prefer to live in warm, shallow water that is 20–29 degrees Celsius ($^{\circ}\text{C}$). The following map shows where coral reefs are located on Earth.

**LEGEND**

■ = Coral reef locations

- 7 The students constructed a Punnett square for corals that illustrates the transmission of the trait that allows the corals to live in warm water.

		Coral Polyp X	
		T	t
Coral Polyp Y	t	Tt	tt
	t	Tt	tt

KEY

T = Dominant
t = Recessive

Coral polyps X and Y are

- Ⓐ the parents and have identical allele pairs.
- Ⓑ the parents and have different allele pairs.
- Ⓒ the offspring and have identical allele pairs.
- Ⓓ the offspring and have different allele pairs.

- Which table best describes the most likely number of chromosomes in coral parents and offspring during sexual reproduction?**

Number of Chromosomes in Each Parent	Number of Chromosomes Provided by Parent 1	Number of Chromosomes Provided by Parent 2	Number of Chromosomes in Each Offspring
28	28	28	28

Number of Chromosomes in Each Parent	Number of Chromosomes Provided by Parent 1	Number of Chromosomes Provided by Parent 2	Number of Chromosomes in Each Offspring
56	28	28	56

Number of Chromosomes in Each Parent	Number of Chromosomes Provided by Parent 1	Number of Chromosomes Provided by Parent 2	Number of Chromosomes in Each Offspring
28	14	14	28

Number of Chromosomes in Each Parent	Number of Chromosomes Provided by Parent 1	Number of Chromosomes Provided by Parent 2	Number of Chromosomes in Each Offspring
7	14	14	28

- 9 The students continued their research to determine how different coral colonies interact with one another.

What method of reproduction do corals use when different colonies reproduce with one another?

- Ⓐ asexual reproduction, which will most likely cause a decrease in potential traits
- Ⓑ sexual reproduction, which will most likely cause an increase in genetic variation
- Ⓒ asexual reproduction, which will most likely cause a decrease in stress from bleaching
- Ⓓ sexual reproduction, which will most likely cause an increase in temperature sensitivity

- 11 The students' research indicated that corals that can survive in water over 31 degrees Celsius ($^{\circ}\text{C}$) have a greater likelihood of surviving bleaching events.**

If coral polyps only breed with corals that carry this temperature trait, the offspring they produce will most likely have

- Ⓐ an increased likelihood of carrying the temperature trait.
- Ⓑ a decreased likelihood of passing on the temperature trait.
- Ⓒ a decreased likelihood of surviving future bleaching events.
- Ⓓ an increased likelihood of experiencing future bleaching events.

-

GENERATION 2

	T	t
t	Tt	tt
t	Tt	tt

	T	t
T	TT	Tt
t	Tt	tt

T = Dominant
t = Recessive

Write your response on the lines provided.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Section 3

Directions:

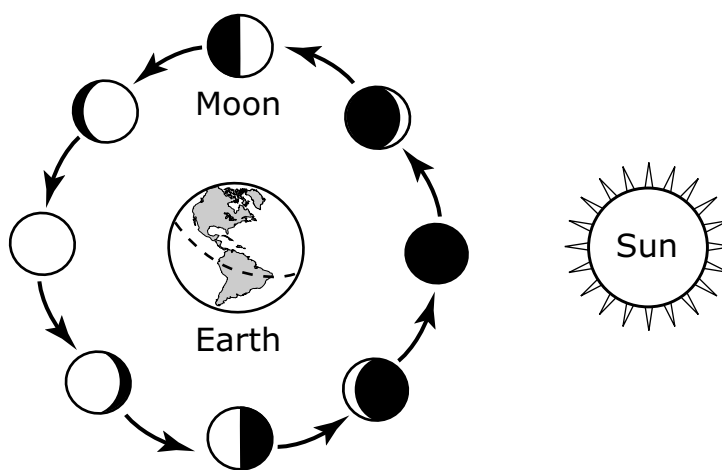
Today, you will take Section 3 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

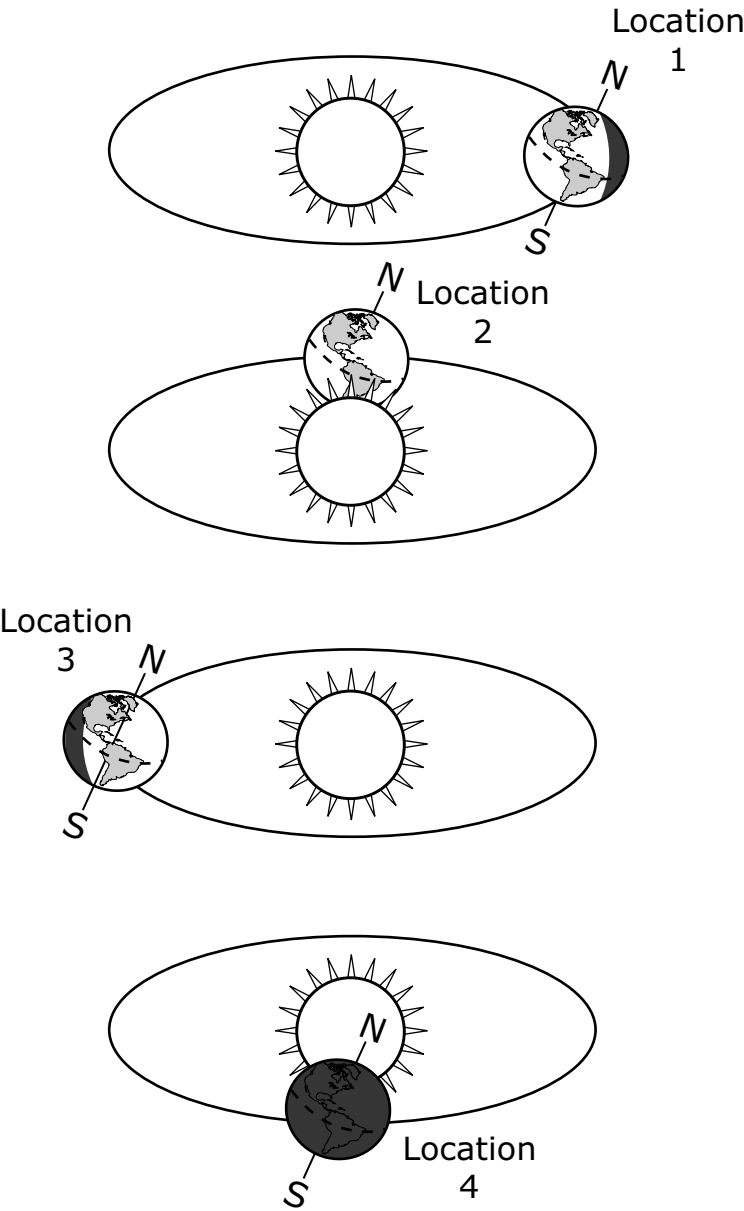
If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

While studying the solar system, students investigated the phases of the moon. They developed a model to demonstrate the phases of the moon. In the model, a golf ball represented the moon, a basketball represented Earth, and a lamp represented the sun. The golf ball was moved in different positions around the basketball, and the students observed how the shadow on the golf ball changed. The students used their observations to construct the following diagram.



The students then researched the effect the sun has on Earth at different times of the year. The students drew the following diagrams showing Earth in four different locations.

DIAGRAM OF SEASONS

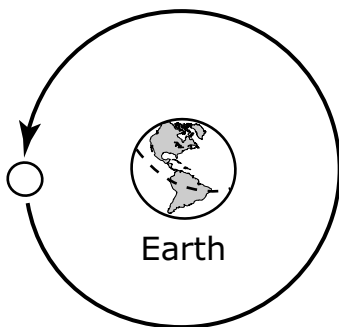


The students continued to research facts about Earth and other celestial objects in the solar system. The students constructed a table to display the data.

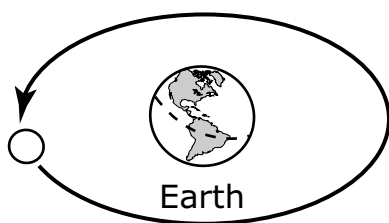
- 1 The students were asked to predict what would happen if the gravitational pull of Earth diminished.

Use the students' model to identify the most likely motion of the moon if the gravitational pull of Earth diminishes.

(A)



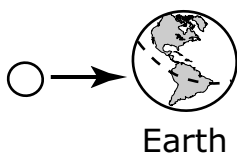
(B)



(C)



(D)



3 Which season is occurring in Maryland when Earth is at Location 1 in the Diagram of Seasons?

- Ⓐ fall
- Ⓑ winter
- Ⓒ spring
- Ⓓ summer

4 When the students researched the planets in the solar system, one student learned that Jupiter has 67 moons.

Jupiter most likely has 67 moons because Jupiter

- Ⓐ is the gaseous planet closest to the sun.
- Ⓑ is the largest celestial object in the solar system.
- Ⓒ has less density and produces a weak gravitational force.
- Ⓓ has a large mass that produces a strong gravitational force.

5 The students constructed another model of the solar system using round objects of various sizes.

If the students chose a volleyball, which is 25.6 centimeters in diameter, as the sun, then the round object the students need for Earth would be approximately

- Ⓐ half the diameter of the volleyball.
- Ⓑ twice the diameter of the volleyball.
- Ⓒ 100 times less than the diameter of the volleyball.
- Ⓓ 100 times more than the diameter of the volleyball.

- Write your response on the lines provided.**

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Read all of the information. Use the information to answer the questions.

While researching information for a science project, several students found a news article from the National Aeronautics and Space Administration (NASA) regarding the Asteroid Redirect Mission (ARM). The focus of this mission is to develop a first-ever robotic mission to visit a large near-Earth asteroid, collect a multi-ton boulder from its surface, and redirect the boulder into a stable orbit around the moon.

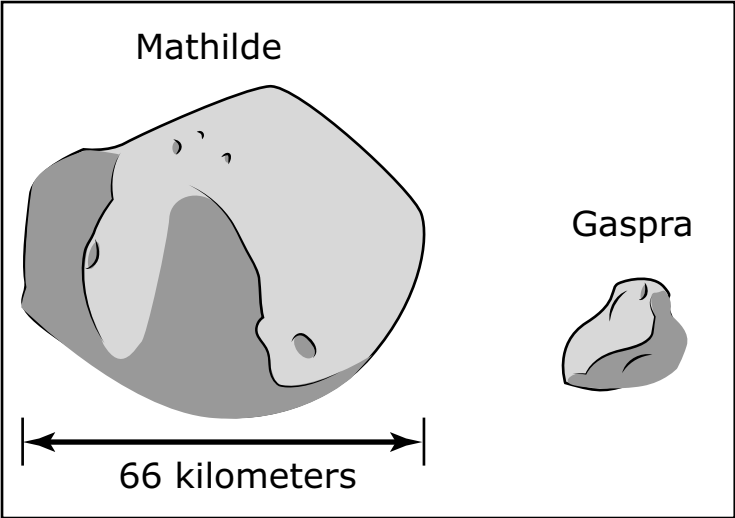
The students continued to research these asteroids and found the following data.

- Asteroids are pieces of rock or metal floating through space.
- In our solar system, there is a large concentration of asteroids in the asteroid belt, an area between Mars and Jupiter.
- Scientists estimate that millions of asteroids are found in this area.
- Some of these asteroids are large, but many are small.
- Scientists think that many asteroids were formed by collisions between other asteroids, moons, and planets.



- **Asteroids in our solar system orbit the sun.**
- **Asteroids can orbit a planet or larger asteroid.**
- **Smaller asteroids do not have enough gravity to pull themselves into a round shape.**
- **More massive asteroids have a more circular shape.**
- **It is possible for the orbits of asteroids and planets to cross, resulting in the chance of a collision.**
- **Scientists have studied past asteroid collisions with Earth and continue to monitor the orbits of asteroids in our solar system.**

The research indicated that asteroids range in size from small rocks to massive boulders that may be hundreds of kilometers wide. The diagram shows two asteroids from our solar system. The table displays data on several other asteroids in the asteroid belt.



ASTEROIDS IN THE ASTEROID BELT

Asteroid Number	Asteroid Name	Diameter (kilometers)	Mass (10^{15} kilograms)
3	Juno	234	20,000
4	Vesta	569	259,000
45	Eugenia	215	6,100
253	Mathilde	66	103.3
951	Gaspra	19	10
4979	Otawara	5.5	0.2
6489	Golevka	1.4	0.00021
25143	Itokawa	0.05	0.000035

A Force with the Power to Move an Asteroid

Could a large asteroid hit Earth? Scientists do not believe that will happen anytime soon, but a new discovery will help them be more certain.

Using powerful radar, scientists tracked the path of an asteroid named 6489 Golevka.

Most asteroids in our solar system travel between the orbits of Mars and Jupiter. Sometimes an asteroid will leave that path and move closer to Earth. Golevka is a near-Earth asteroid, which means that it comes within 121 million miles (195 million kilometers) of the sun.

Using radar, scientists found that Golevka's path moved 9.4 miles (15 kilometers) in the last 12 years. That distance may seem small compared with the billions of miles the asteroid traveled during that time, but Steven Chesley, a NASA scientist and leader of a new asteroid study, believes that those ten miles are important.

"Over tens of millions of years that [change] can have a big effect," he explained.

But what caused asteroid Golevka's path to change?

Scientists say that a force called the Yarkovsky effect moved the asteroid. The effect happens when the sun heats one part of an asteroid's surface more than other parts. The uneven heating causes a force that pushes an asteroid out of its normal path.

"We measured a force of about one ounce (28 grams¹) acting on an asteroid that weighs 460 billion pounds (208 billion kilograms)" Chesley explained. That means that a force equal to about the weight of a strawberry can change the course of an asteroid that is longer than five football fields!

While no large asteroids appear on track to hit Earth, the Yarkovsky effect's ability to predict asteroids' paths could help scientists be more certain and give them some of the information necessary to stop a collision.

"A Force with the Power to Move an Asteroid" – Sarah Ives, © 2004, nationalgeographic.com

¹**28 grams:** approximately 0.27 newton

7 After researching the orbits of asteroids in the solar system, students explained that in order for asteroids in the asteroid belt to remain in orbit,

- Ⓐ Earth must exert a strong gravitational force toward the center of the solar system.
- Ⓑ the sun must exert a strong gravitational force toward the center of the solar system.
- Ⓒ Earth must exert a strong gravitational force away from the center of the solar system.
- Ⓓ the sun must exert a strong gravitational force away from the center of the solar system.

8 The students used evidence to present an argument that scientists should launch an unmanned spacecraft to prevent a collision between an asteroid and Earth's moon.

In order for the spacecraft to prevent the asteroid from colliding with the moon, the spacecraft would most likely

- Ⓐ have a smaller mass than the mass of the asteroid.
- Ⓑ have enough force to knock the asteroid off course.
- Ⓒ reduce its speed to match the speed of the asteroid.
- Ⓓ spin in a forward direction to alter the orbit of the asteroid.

- A**

(B)

©

Ⓓ

SERIAL #

10 Part A

Data from the students' research indicated that some asteroids are orbited by other asteroids.

Which asteroid would most likely be orbited by other asteroids?

- Ⓐ Itokawa
- Ⓑ Juno
- Ⓒ Mathilde
- Ⓓ Vesta

Part B

The explanation that best describes why one asteroid would orbit another asteroid is that the asteroid being orbited has a large

- Ⓐ mass, which results in a strong gravitational force.
- Ⓑ metallic core, which results in a strong magnetic field.
- Ⓒ diameter, which results in a strong gravitational force.
- Ⓓ amount of iron, which results in a strong magnetic field.

-



- 57

12 Asteroids orbit other asteroids similarly to moons orbiting a planet. The following table identifies the mass of Earth and Jupiter and the number of moons for each planet.

MASS OF CELESTIAL OBJECTS

Celestial Object	Mass (10 ²⁴ kilograms)	Number of Moons
Earth	5.97	1
Jupiter	1898.00	67

Write your response on the lines provided.

[illegible]

Section 4

Directions:

Today, you will take Section 4 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

- If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section **ONLY**. Do not go past the stop sign.

Read all of the information. Use the information to answer the questions.

What do asphalt, heating oil, and scented soaps have in common? Some varieties of these products are made from crude oil. Crude oil, also known as petroleum, is a resource that is extracted from the earth and refined into various products. Some companies continue to use crude oil more than healthier options in developing their products because it saves them money.

A student studied alternative ways to develop soaps that are healthier for people and safer for the environment, such as using coconut oil to make soap. The student obtained information on how to create soap from an online resource and asked the teacher to repeat the process in the classroom. The teacher wore goggles and followed proper lab safety procedures when making the soap.

Making Soap from Coconut Oil

Materials

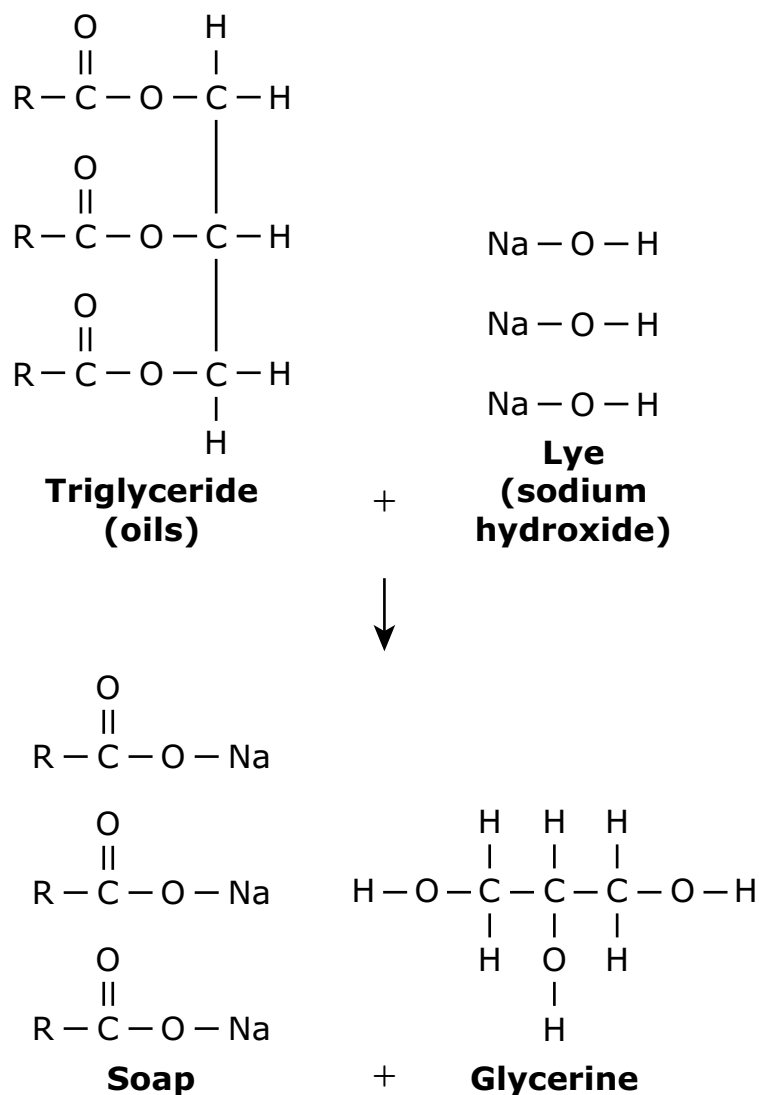
- coconut oil
- castor oil
- lye (sodium hydroxide)
- water

Steps

1. Melt the coconut oil in a pan.
2. Add the lye to the water and mix to dissolve. The mixture will become warm. Let cool to room temperature.
3. Pour the castor oil and melted coconut oil into a large pot. Let the mixture cool to room temperature.
4. Slowly add the lye solution to the oils and mix them together. This combined solution will be warm.
5. Continue mixing until the solution has a thin pudding-like consistency. This is the time to add color and fragrance (scent) if you decide to use them.
6. Pour the mixture into containers with different shapes for the soap.
7. Let the finished soap solidify overnight.

[illegible]

The student created a general model to show the chemical reaction for making soap.



KEY

C — Carbon
 O — Oxygen
 H — Hydrogen
 Na — Sodium
 R — R group (other atoms not shown)

- 65

PLEASE DO NOT WRITE IN THIS AREA

SERIAL #

3 Which step in the procedure can be used as evidence to support the claim that chemical reactions can turn natural resources into synthetic materials?

- Ⓐ Step 1
- Ⓑ Step 5
- Ⓒ Step 6
- Ⓓ Step 7

4 How many different types of reactant molecules are used in the chemical reaction to make soap?

- Ⓐ 2
- Ⓑ 4
- Ⓒ 5
- Ⓓ 8

5 Which statement best describes the types of atoms in the soap reaction?

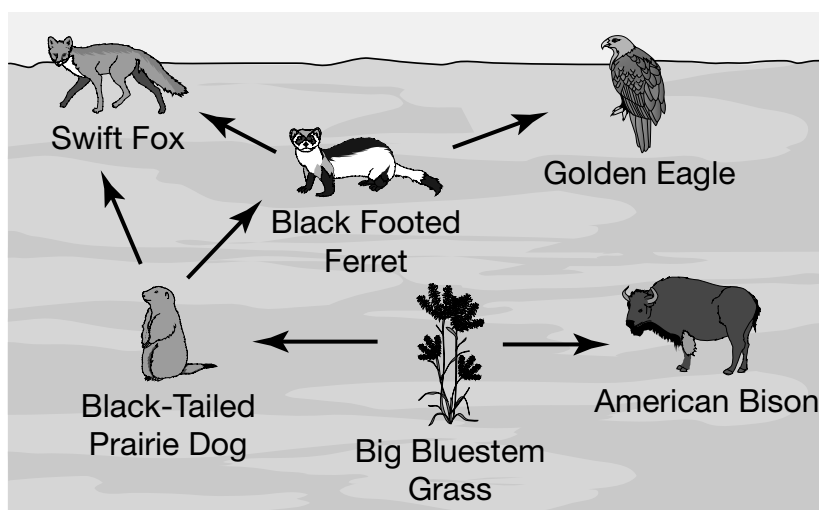
- Ⓐ All of the atoms have the same mass as one another.
- Ⓑ Each of the atoms has a specific mass, which is the same for all atoms of that type.
- Ⓒ Each of the atoms appears in every reactant molecule in the soap reaction.
- Ⓓ Each of the atoms appears in every product molecule in the soap reaction.

Write your response on the lines provided.

[illegible]

Read all of the information. Use the information to answer the questions.

During a visit to the Maryland Zoo, a group of students observed a colony of black-tailed prairie dogs. The students decided to research prairie dogs and the North American prairie ecosystem in which they live. The students found the following food web that illustrated relationships between several organisms in the prairie ecosystem.



INITIAL FERRET POPULATION 20

Species	Initial Population	Population after 2 years	Population after 4 years	Population after 6 years	Population after 8 years
Grass (kilograms)	4000	2000	500	1000	5000
Prairie Dogs	25000	31000	8000	4000	11000
Ferrets	20	25	35	15	15
Foxes	10	4	4	3	2

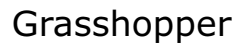
INITIAL FERRET POPULATION 80

Species	Initial Population	Population after 2 years	Population after 4 years	Population after 6 years	Population after 8 years
Grass (kilograms)	4000	4000	4000	4000	4000
Prairie Dogs	25000	25000	25000	25000	25000
Ferrets	80	80	80	80	80
Foxes	10	10	10	10	10

INITIAL FERRET POPULATION 140

Species	Initial Population	Population after 2 years	Population after 4 years	Population after 6 years	Population after 8 years
Grass (kilograms)	4000	5100	4900	2800	3900
Prairie Dogs	25000	21000	28000	26000	22500
Ferrets	140	75	75	90	75
Foxes	10	11	10	10	10

SPECIES THAT HAVE RELATIONSHIPS WITH PRAIRIE DOGS



- ## American Bison

- eat the same type of grass as the prairie dogs
- eat the taller grass that has less nutrients than the shorter grass
- fertilize the soil with dung which helps grass grow
- roll in the dirt mounds created by prairie dogs digging tunnels to help keep biting flies away
- produce one calf each year



Burrowing Owl

- live underground in burrows that have been dug out and abandoned by prairie dogs
- eat grasshoppers
- produce **3-12** hatchlings a year
- are active in the daytime, unlike other types of owls
- may collect bison dung around burrows when nesting

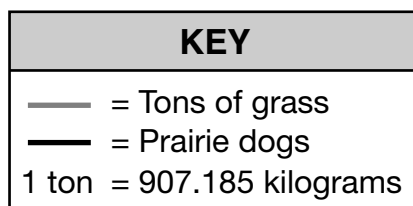
- 8 The prairie food web diagram illustrates interactions among organisms that live in the prairie ecosystem.**

Which statement best describes the sources of energy for the producers and consumers in the food web?

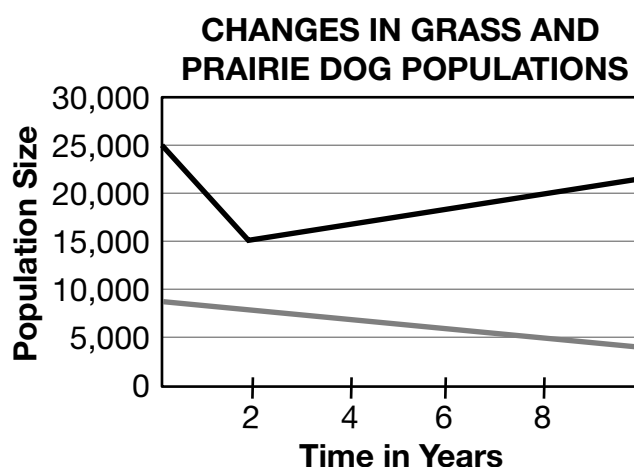
- Ⓐ Consumers and producers both obtain energy from decomposers.
- Ⓑ Consumers gain energy from the sun, while producers obtain energy by eating other organisms.
- Ⓒ Producers obtain energy from living organisms, while consumers obtain energy from the nonliving parts of the ecosystem.
- Ⓓ Producers use the sun and nonliving parts of the ecosystem to generate energy, while consumers gain energy from other living organisms.

- 10 The research stated that the prairie dog and ferret interact with one another in the prairie ecosystem.

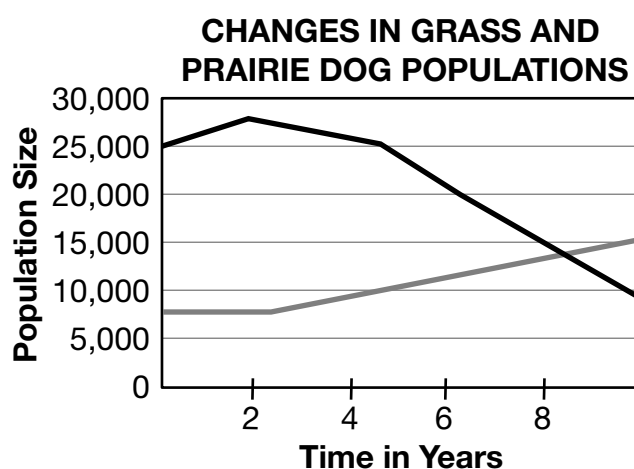
Which graph best represents the changes in the amount of grass and prairie dog population when the initial number of ferrets in the ecosystem was 140?



(A)



(B)



CHANGES IN GRASS AND PRAIRIE DOG POPULATIONS



CHANGES IN GRASS AND PRAIRIE DOG POPULATIONS



- 12 The three prairie food web resources illustrate the interactions among organisms in the prairie ecosystem.**

Describe how the prairie dog and bison populations will most likely change if a severe drought were to occur for a four-year period. In your description, be sure to include

- the cause of any changes to the populations
- the movement of energy within the ecosystem
- the interactions among the organisms

Write your response on the lines provided.

[illegible]

The following pages include the answer key for all machine-scored items. Refer to the separate documents for all hand-scored items as indicated in the answer key.

Section 1

Item Number	Answer Key	Performance Expectation
1.	A	MS-LS1-4
2.	B, C	MS-LS1-4
3.	A	MS-LS1-5
4.	A	MS-LS1-5
5.	B	MS-LS1-5
6.	Refer to Maryland Integrated Science Assessment Rubrics	MS-LS1-4
7.	B	MS-ESS3-5
8.	A, E	MS-ESS3-5
9.	C	MS-ESS1-4

Table continues on next page.

Section 1, continued

Item Number	Answer Key	Performance Expectation
10.	C	MS-ESS1-4
11.	B	MS-ESS1-4
12.	Refer to Maryland Integrated Science Assessment Rubrics	MS-ESS3-5

Section 2

Item Number	Answer Key	Performance Expectation
1.	C	MS-ETS1-1
2.	D	MS-ETS1-1
3.	B, C	MS-ETS1-1
4.	A	MS-ETS1-1
5.	C	MS-PS4-2
6.	Refer to Maryland Integrated Science Assessment Rubrics	MS-PS4-2
7.	B	MS-LS3-2
8.	C	MS-LS3-2
9.	B	MS-LS4-4
10.	Part A: D Part B: B, C, E	MS-LS4-4
11.	A	MS-LS4-4
12.	Refer to Maryland Integrated Science Assessment Rubrics	MS-LS3-2

Section 3

Item Number	Answer Key	Performance Expectation
1.	C	MS-ESS1-2
2.	C, D	MS-ESS1-1
3.	B	MS-ESS1-1
4.	D	MS-ESS1-2
5.	C	MS-ESS1-3
6.	Refer to Maryland Integrated Science Assessment Rubrics	MS-ESS1-3
7.	B	MS-PS2-4
8.	B	MS-PS2-1
9.	D	MS-PS2-4
10.	Part A: D Part B: A	MS-PS2-4
11.	A	MS-PS2-4
12.	Refer to Maryland Integrated Science Assessment Rubrics	MS-PS2-4

Section 4

Item Number	Answer Key	Performance Expectation
1.	B	MS-PS1-3
2.	D	MS-PS1-3
3.	B	MS-PS1-3
4.	A	MS-PS1-5
5.	B	MS-PS1-5
6.	Refer to Maryland Integrated Science Assessment Rubrics	MS-PS1-5
7.	A	MS-LS2-3
8.	D	MS-LS2-3
9.	B	MS-LS2-3
10.	D	MS-LS2-1
11.	A	MS-LS2-2
12.	Refer to Maryland Integrated Science Assessment Rubrics	MS-LS2-1